The Sampling Distribution of PRE and the F-Distribution.

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The sampling distribution of PRE

- In class, we will show the sampling distribution of PRE using simulations.

- For now, let's focus on two key concepts:
  - The proportional reduction in error will always be ≥ 0 in this situation, because the sample mean will provide the smallest SSE for a set of numbers. (If the mean had been 37 exactly, PRE would be 0.)

  - The larger the proportional reduction in error, the less likely we think a result is to have occurred from sampling variability alone. (That is, if the variability explained by the estimate is large compared to the residual variability, there might be something ***systematic*** rather than ***random*** happening in our data.)

The sampling distribution of PRE

- "Know thy distributions."

- In order for us to make an inferential statement about the probability of observing PRE, we need to make an estimate about the variance in the population.

- As we have shown, the sample variance is an unbiased estimate of the population variance:
  - $\sigma^2 \approx s^2 = \frac{SSE}{(n-1)} = \frac{\sum(y_i - \bar{y})^2}{(n-1)}$

The sampling distribution of PRE

- Critically, the shape of the sampling distribution depends on the number of estimated parameters in Model A (PA) and the number of observations in the sample (n).
  - $\sigma^2 \approx s^2 = \frac{SSE}{(n-1)} = \frac{\sum(y_i - \bar{y})^2}{(n-1)}$

- We are estimating one parameter (the mean) and we have 10 observations.
  - As the number of observations increases, s^2 will be a more consistent approximation of σ^2 and thus a larger PRE is more surprising.
  - As the number of parameters increases, we expect to explain more error (even just by chance) and thus a larger PRE is less surprising.

The sampling distribution of PRE

- Thus, the number of parameters in Model A (PA) and the number of observations in the sample (n) work against each other in how we interpret the PRE we observe.

- However, critical values of PRE and its distribution are rare in statistics books. Instead, it is much more common to see PRE converted into an ***F-value***, and interpreted with respect to the ***F-distribution***.

The F-statistic and F-distribution

- The F-statistic is defined as:

$$F = \frac{PRE/(PA - PC)}{(1 - PRE)/(n - PA)}$$

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- Where $\frac{PRE}{(PA-PC)}$ is proportional reduction in error, per parameter, for the comparison of Model A to Model C.
- And $\frac{(1-PRE)}{(n-PA)}$ is the remaining, unexplained error per the remaining degrees of freedom.

The F-statistic and F-distribution

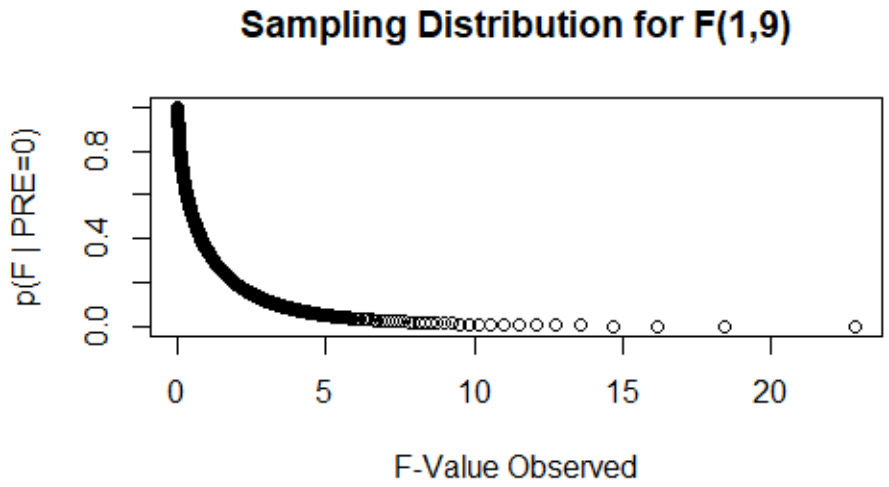
- The F-statistic is defined as:

$$F = \frac{PRE/(PA - PC)}{(1 - PRE)/(n - PA)}$$

- Thus, the F-value can be interpreted as the ratio of explained variance to unexplained variance, taking into account the sample size and number of parameters.

The F-statistic and F-distribution

- Our particular F-observed value can then be compared to the F-distribution for that number of parameters (df1) and number of observations (df2).



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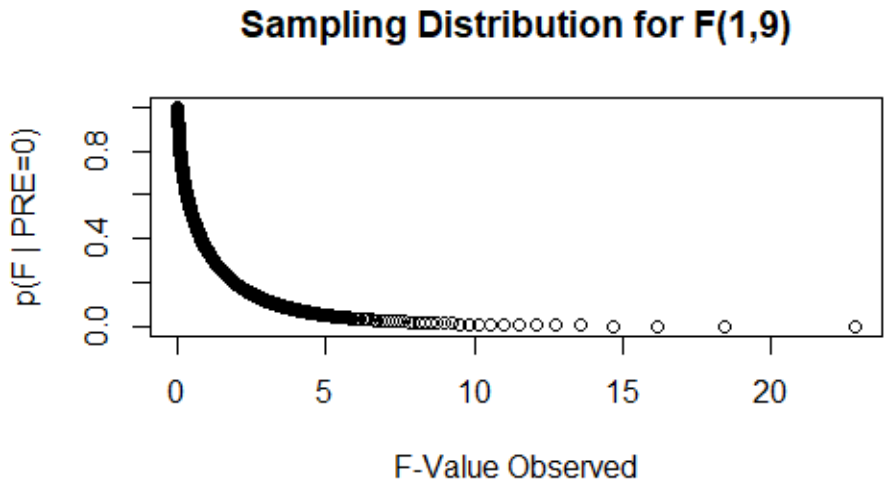

$$F = \frac{0.603/(PA - PC)}{(1 - 0.603)/(n - PA)}$$

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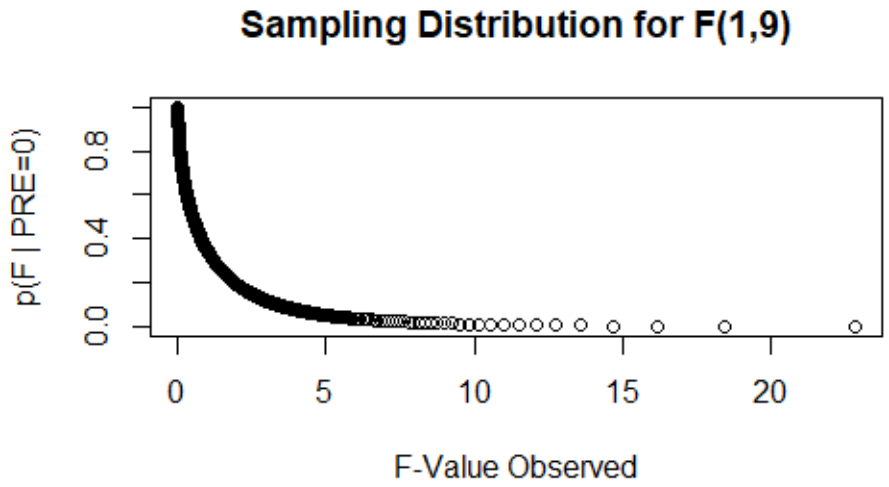

$$F = \frac{0.603/(1-0)}{(1-0.603)/(10-1)}$$

The F-statistic and F-distribution

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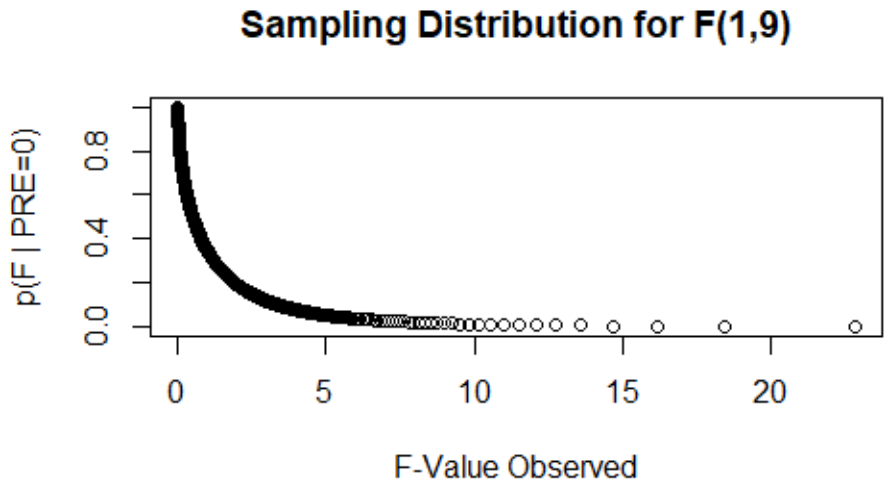

$$F = \frac{0.603/(1-0)}{(1-0.603)/(10-1)}$$

$$F = \frac{0.603}{(0.397)/(9)}$$

$$F = 13.67$$

The F-statistic and F-distribution

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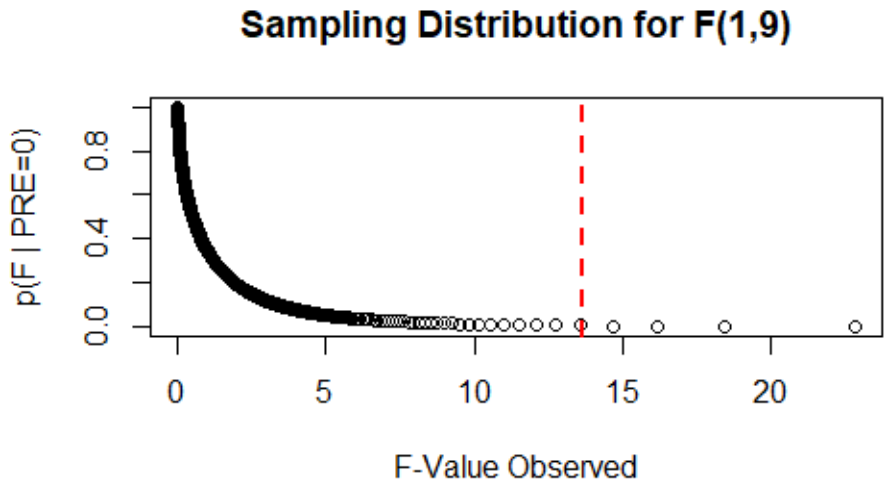

$$F = 13.67$$

Assuming the null-hypothesis, with 1 and 9 degrees of freedom, we would expect to see an F-value ≥ 13.67 about 0.005% of the time due to sampling variability alone.

F(1,9) = 13.67, p = 0.005